AWS CLOUD & DATA ENGINEERING ASSIGNMENT

Name : Prajna BR

Roll Number : 23BTRIS028

College Name : Jain University

Batch : 2027

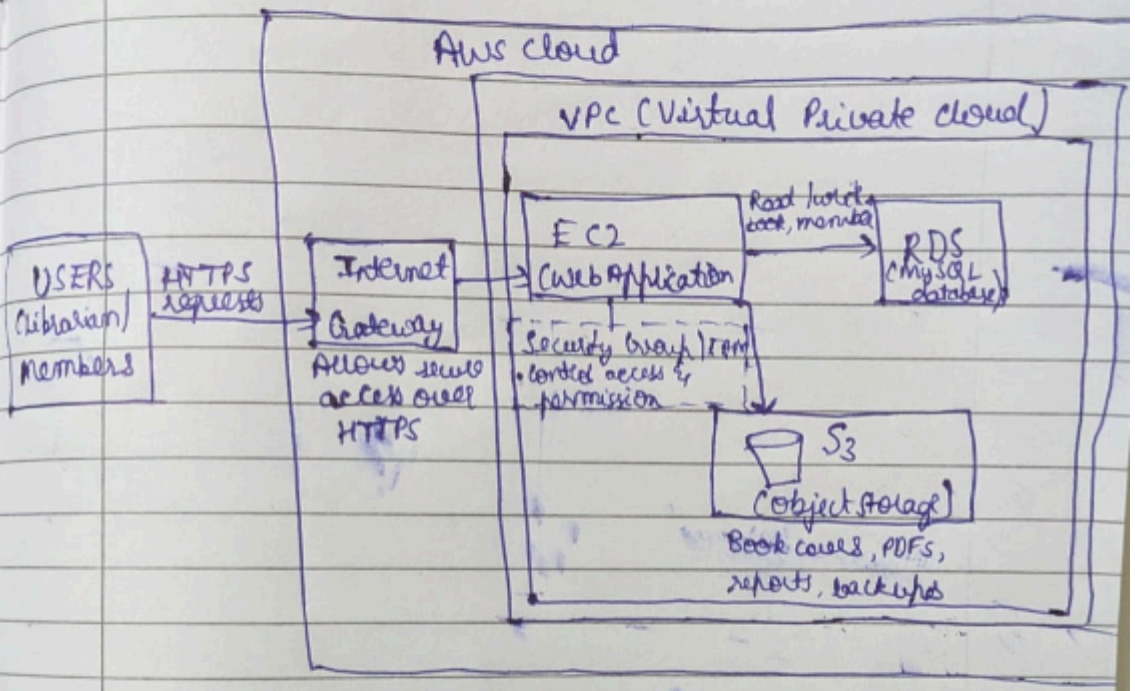
Date : 10-10-2026

Title : AWS Cloud & Data Engineering

Library Management System
AWS Cloud Architecture

URBAN EDGE

- A cloud based system to manage books, members, borrowing, returns & library records



- System Architecture components:

→ Amazon EC_2 : Run the library web application

→ Amazon RDS : Stores books, members & borrowing/ return records.

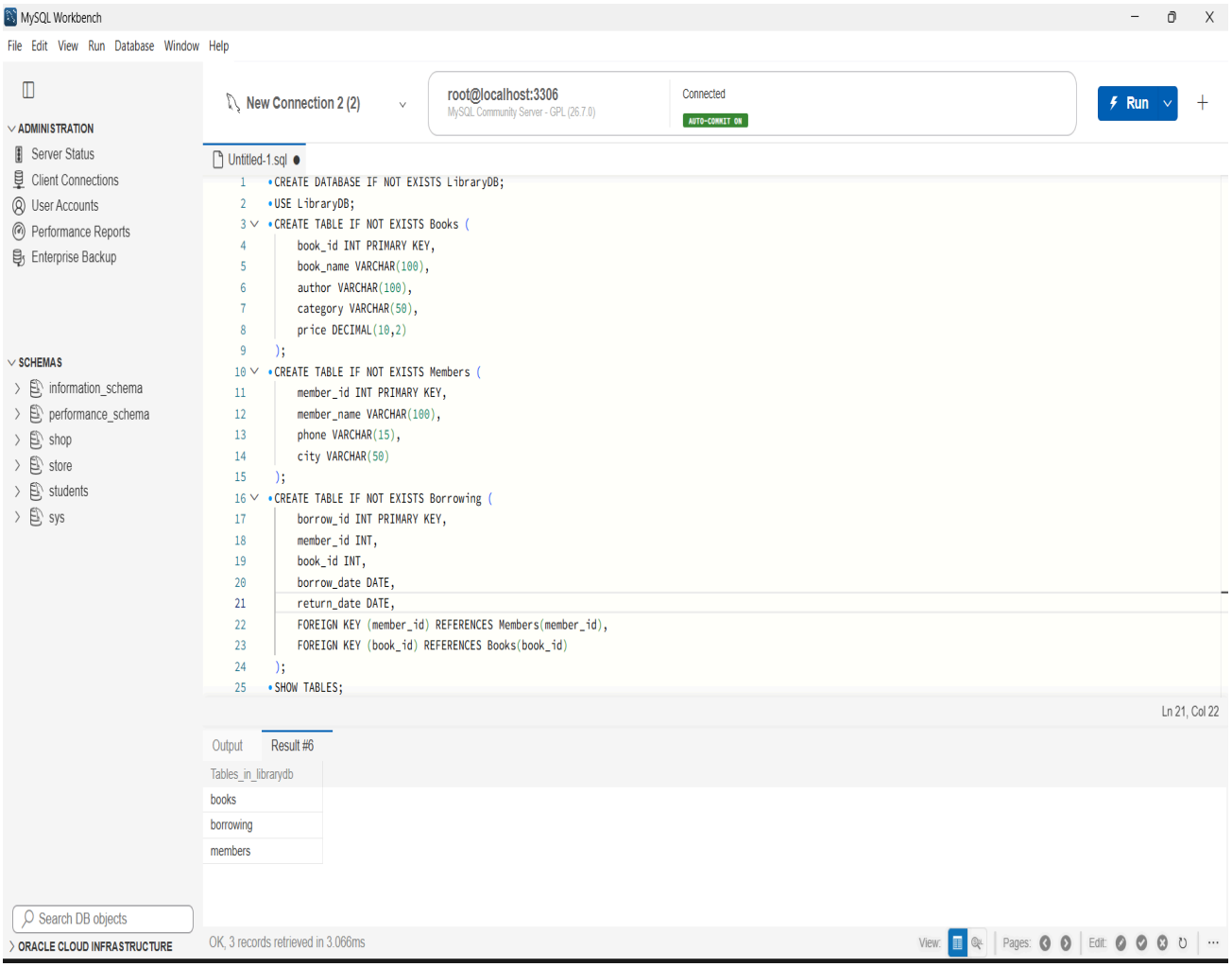
→ Amazon S3 : Stores book covers images, PDFs & files

→ Security Groups/IAM : Restrict network access & user permissions.

→ Data flow : User → HTTPS → EC_2 → RDS / S3

SQL

1. Create a Table in MySQL

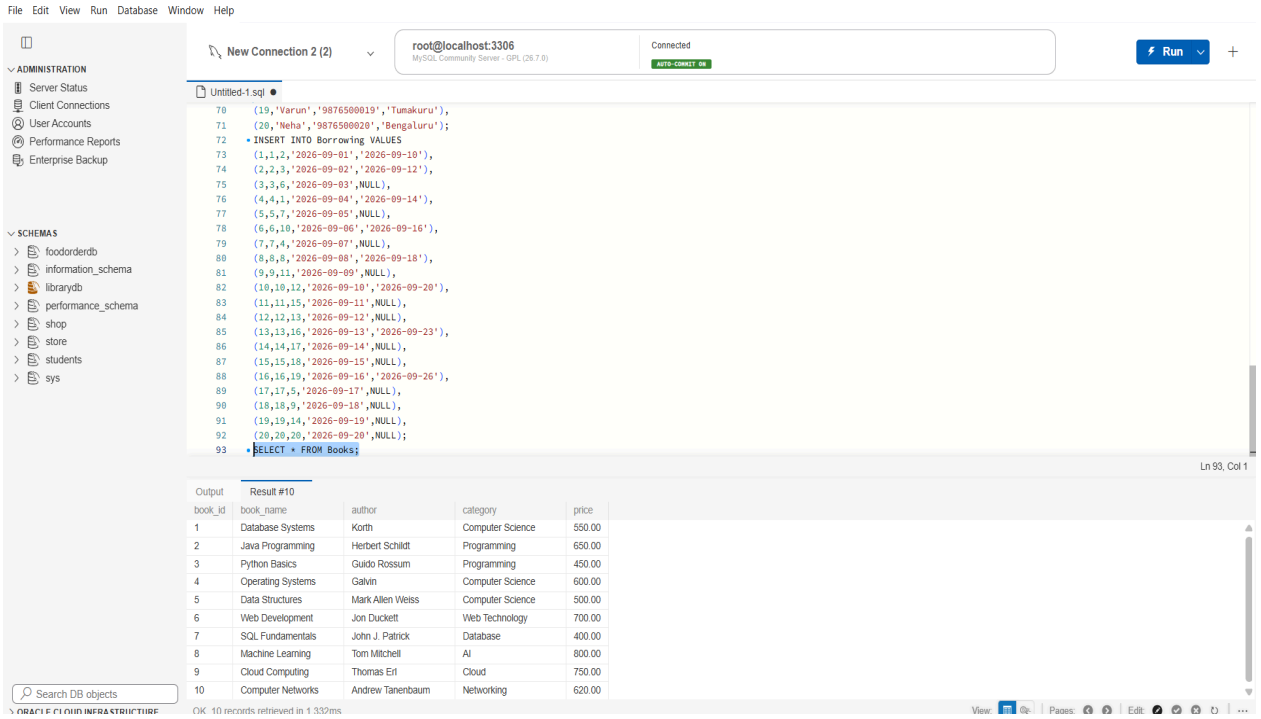


2. Inserting Values

```
INSERT INTO Books VALUES
(1,'Wings of Fire','A.P.J. Abdul Kalam','Biography',350),
(2,'The Alchemist','Paulo Coelho','Fiction',400),
(3,'Atomic Habits','James Clear','Self Help',550),
(4,'Rich Dad Poor Dad','Robert Kiyosaki','Finance',450),
(5,'The Great Gatsby','F. Scott Fitzgerald','Fiction',300),
(6,'Clean Code','Robert Martin','Technology',750),
(7,'Java Programming','Herbert Schildt','Technology',650),
(8,'Python Crash Course','Eric Matthes','Technology',700),
(9,'Think and Grow Rich','Napoleon Hill','Finance',380),
(10,'Ikigai','Hector Garcia','Self Help',320),
(11,'1984','George Orwell','Fiction',280),
(12,'Deep Work','Cal Newport','Productivity',500),
(13,'Introduction to Algorithms','Thomas Cormen','Technology',950),
(14,'The Power of Now','Eckhart Tolle','Self Help',420),
(15,'Sapiens','Yuval Noah Harari','History',600),
(16,'Database System Concepts','Abraham Silberschatz','Technology',850),
(17,'The Hobbit','J.R.R. Tolkien','Fantasy',390),
(18,'The Psychology of Money','Morgan Housel','Finance',488),
(19,'Harry Potter','J.K. Rowling','Fantasy',520),
(20,'Pride and Prejudice','Jane Austen','Fiction',348);
INSERT INTO Members VALUES
(1,'Aarav','9876500001','Bengaluru'),
(2,'Ananya','9876500002','Mysuru'),
(3,'Rahul','9876500003','Bengaluru'),
(4,'Priya','9876500004','Mangaluru'),
(5,'Arjun','9876500005','Hubballi'),
(6,'Sneha','9876500006','Bengaluru'),
(7,'Kiran','9876500007','Mysuru'),
(8,'Divya','9876500008','Shivamogga'),
(9,'Rohan','9876500009','Bengaluru'),
(10,'Pooja','9876500010','Tumakuru'),
(11,'Vikram','9876500011','Mangaluru'),
(12,'Meera','9876500012','Bengaluru'),
(13,'Aditya','9876500013','Mysuru'),
(14,'Kavya','9876500014','Udupi'),
(15,'Nikhil','9876500015','Hubballi'),
(16,'Sanjana','9876500016','Bengaluru'),
(17,'Manoj','9876500017','Belagavi'),
(18,'Aishwarya','9876500018','Mysuru'),
(19,'Varun','9876500019','Tumakuru'),
(20,'Neha','9876500020','Bengaluru');
INSERT INTO Borrowing VALUES
(1,1,2,'2026-09-01','2026-09-10'),
(2,2,3,'2026-09-02','2026-09-12'),
(3,3,6,'2026-09-03',NULL),
(4,4,1,'2026-09-04','2026-09-14'),
(5,5,7,'2026-09-05',NULL),
(6,6,18,'2026-09-06','2026-09-16'),
(7,7,4,'2026-09-07',NULL),
(8,8,8,'2026-09-08','2026-09-18'),
(9,9,11,'2026-09-09',NULL),
(10,10,12,'2026-09-10','2026-09-20'),
(11,11,15,'2026-09-11',NULL),
(12,12,13,'2026-09-12',NULL),
(13,13,16,'2026-09-13','2026-09-23'),
(14,14,17,'2026-09-14',NULL),
(15,15,18,'2026-09-15',NULL),
(16,16,19,'2026-09-16','2026-09-26'),
(17,17,5,'2026-09-17',NULL),
(18,18,9,'2026-09-18',NULL),
(19,19,14,'2026-09-19',NULL),
(20,20,20,'2026-09-20',NULL);
```

SQL Quires

1. Display all books



2. Display all members

```
91    (19,19,14,'2026-09-19',NULL),
92    (20,20,20,'2026-09-20',NULL);
93  • SELECT * FROM Books;
94  • SELECT * FROM Members;
```

Output | Result #11

member_id	member_name	phone	city
1	Rahul	9876543210	Bengaluru
2	Priya	9876543211	Mysuru
3	Arjun	9876543212	Mangaluru
4	Sneha	9876543213	Bengaluru
5	Kiran	9876543214	Hubballi
6	Ananya	9876543215	Mysuru
7	Vikram	9876543216	Bengaluru
8	Divya	9876543217	Udupi
9	Rohit	9876543218	Mangaluru
10	Meena	9876543219	Bengaluru

OK, 10 records retrieved in 1.73ms

3.Display all borrowing records

```
93  SELECT * FROM Books;
94  SELECT * FROM Members;
95  SELECT * FROM Borrowing;
```

Output | Result #12

borrow_id	member_id	book_id	borrow_date	return_date
2	2	2	02/10/2026	16/10/2026
3	3	4	03/10/2026	17/10/2026
4	4	3	04/10/2026	18/10/2026
5	5	5	05/10/2026	19/10/2026
6	1	6	06/10/2026	20/10/2026
7	6	7	07/10/2026	21/10/2026
8	7	8	08/10/2026	22/10/2026
9	8	9	09/10/2026	23/10/2026
10	9	10	10/10/2026	24/10/2026
11	2	1	11/10/2026	25/10/2026
12	3	5	12/10/2026	26/10/2026
13	4	2	13/10/2026	27/10/2026
14	5	4	14/10/2026	28/10/2026
15	10	3	15/10/2026	29/10/2026

OK, 15 records retrieved in 1.509ms

4. Display book names and authors

```
96  SELECT book_name, author
97  FROM Books;
```

Output | Result #13

book_name	author
Database Systems	Korth
Java Programming	Herbert Schildt
Python Basics	Guido Rossum
Operating Systems	Galvin
Data Structures	Mark Allen Weiss
Web Development	Jon Duckett
SQL Fundamentals	John J. Patrick
Machine Learning	Tom Mitchell
Cloud Computing	Thomas Erl
Computer Networks	Andrew Tanenbaum

5. Find books costing more than Rs.5000

```
97  FROM Books;
98  SELECT *
99  FROM Books
100 WHERE price > 500;
```

Output | Result #14

book_id	book_name	author	category	price
1	Database Systems	Korth	Computer Science	550.00
2	Java Programming	Herbert Schildt	Programming	650.00
4	Operating Systems	Galvin	Computer Science	600.00
6	Web Development	Jon Duckett	Web Technology	700.00
8	Machine Learning	Tom Mitchell	AI	800.00
9	Cloud Computing	Thomas Erl	Cloud	750.00
10	Computer Networks	Andrew Tanenbaum	Networking	620.00

6. Display books in the Fiction category

```
101 SELECT *
102 FROM Books
103 WHERE category = 'Fiction';
```

Result #15

book_id	book_name	author	category	price

7.Sort books by price in ascending order

```
104 SELECT *
105 FROM Books
106 ORDER BY price ASC;
```

Output | Result #16

book_id	book_name	author	category
7	SQL Fundamentals	John J. Patrick	Database
3	Python Basics	Guido Rossum	Programming
5	Data Structures	Mark Allen Weiss	Computer Science
1	Database Systems	Korth	Computer Science
4	Operating Systems	Galvin	Computer Science
10	Computer Networks	Andrew Tanenbaum	Networking
2	Java Programming	Herbert Schildt	Programming
6	Web Development	Jon Duckett	Web Technology
9	Cloud Computing	Thomas Erl	Cloud
8	Machine Learning	Tom Mitchell	AI

8. Sort books by price in descending order

```
107    SELECT *
108    FROM Books
109    ORDER BY price DESC;
```

Output | Result #17

book_id	book_name	author	category	price
8	Machine Learning	Tom Mitchell	AI	800.00
9	Cloud Computing	Thomas Erl	Cloud	750.00
6	Web Development	Jon Duckett	Web Technology	700.00
2	Java Programming	Herbert Schildt	Programming	650.00
10	Computer Networks	Andrew Tanenbaum	Networking	620.00
4	Operating Systems	Galvin	Computer Science	600.00
1	Database Systems	Korth	Computer Science	550.00
5	Data Structures	Mark Allen Weiss	Computer Science	500.00
3	Python Basics	Guido Rossum	Programming	450.00
7	SQL Fundamentals	John J. Patrick	Database	400.00

9.Display unique member cities

```
110    SELECT DISTINCT city
111    FROM Members;
```

Output | Result #18

city
Bengaluru
Mysuru
Mangaluru
Hubballi
Udupi

10. Count the total number of books

```
112    SELECT COUNT(*) AS total_books
113    FROM Books;
```

Output | Result #19

total_books
10

11.Find the highest book price

```
114    SELECT MAX(price) AS highest_price
115    FROM Books;
```

Output | Result #20

highest_price
800.00

12. Calculate the average book price

```
115    FROM Books;
116    SELECT AVG(price) AS average_price
117    FROM Books;
```

Output | Result #21

average_price
602.000000

13.Count members in each city

```
118    SELECT city, COUNT(*) AS total_members
119    FROM Members
120    GROUP BY city;
```

Output | Result #22

city	total_members
Bengaluru	4
Mysuru	2
Mangaluru	2
Hubballi	1
Udupi	1

14.Display member names and their borrowed books

```
121    SELECT
122        m.member_name,
123        b.book_name,
124        br.borrow_date
125    FROM Borrowing br
126    JOIN Members m
127        ON br.member_id = m.member_id
128    JOIN Books b
129        ON br.book_id = b.book_id;
```

Output | Result #23

member_name	book_name	borrow_date
Rahul	Database Systems	01/10/2026
Priya	Java Programming	02/10/2026
Arjun	Operating Systems	03/10/2026
Sneha	Python Basics	04/10/2026
Kiran	Data Structures	05/10/2026
Rahul	Web Development	06/10/2026
Ananya	SQL Fundamentals	07/10/2026
Vikram	Machine Learning	08/10/2026
Divya	Cloud Computing	09/10/2026
Rohit	Computer Networks	10/10/2026

OK, 15 records retrieved in 2.782ms

15. Count the number of times each book was borrowed

```
130     SELECT
131     b.book_name,
132     COUNT(br.borrow_id) AS times_borrowed
133 FROM Books b
134 LEFT JOIN Borrowing br
135     ON b.book_id = br.book_id
136 GROUP BY b.book_id, b.book_name
137 ORDER BY times_borrowed DESC;
```

Output | Result #24

book_name	times_borrowed
Database Systems	2
Java Programming	2
Python Basics	2
Operating Systems	2
Data Structures	2
Web Development	1
SQL Fundamentals	1
Machine Learning	1
Cloud Computing	1
Computer Networks	1

Linux Commands

1. Print working directory & List files

```
/mnt/c/Users/brrit
prajna_br@Rithu:/mnt/c/Users/brrit$ ls
 AppData                                                
```

4.Change directory

```
prajna_br@Rithu:/mnt/c/Users/brrit$ cd linux_assignment
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$ pwd
/mnt/c/Users/brrit/linux_assignment
```

5.Write a file & Display file contains

```
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$ echo "Linux Practice"
Linux Practice
```

6.Create a file

```
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$ touch sample.txt
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$ ls -l
total 0
-rwxrwxrwx 1 root root 0 Oct 11 16:45 sample.txt
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$
```

7.Copy file

```
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$ cp sample.txt copy.txt
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$ ls
copy.txt  sample.txt
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$
```

8.Rename a file

```
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$ mv copy.txt renamed.txt
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$ whoami
prajna_br
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$
```

9. Display system information

```
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$ uname -a
Linux Rithu 6.18.40.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Fri Jul 31 22:12:15 UTC 2026 x86_64 GNU/Linux
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$
```

10. Check memory usage

```
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.7Gi       451Mi       3.0Gi       4.0Mi       281Mi       3.2Gi
Swap:          1.0Gi          0B       1.0Gi
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$
```

11. Display date

```
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$ date
Sun Oct 11 16:54:57 UTC 2026
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$
```

12.Display hostname

```
prajna_br@Rithu:/mnt/c/Users/brrit$ hostname
Rithu
prajna_br@Rithu:/mnt/c/Users/brrit$
```

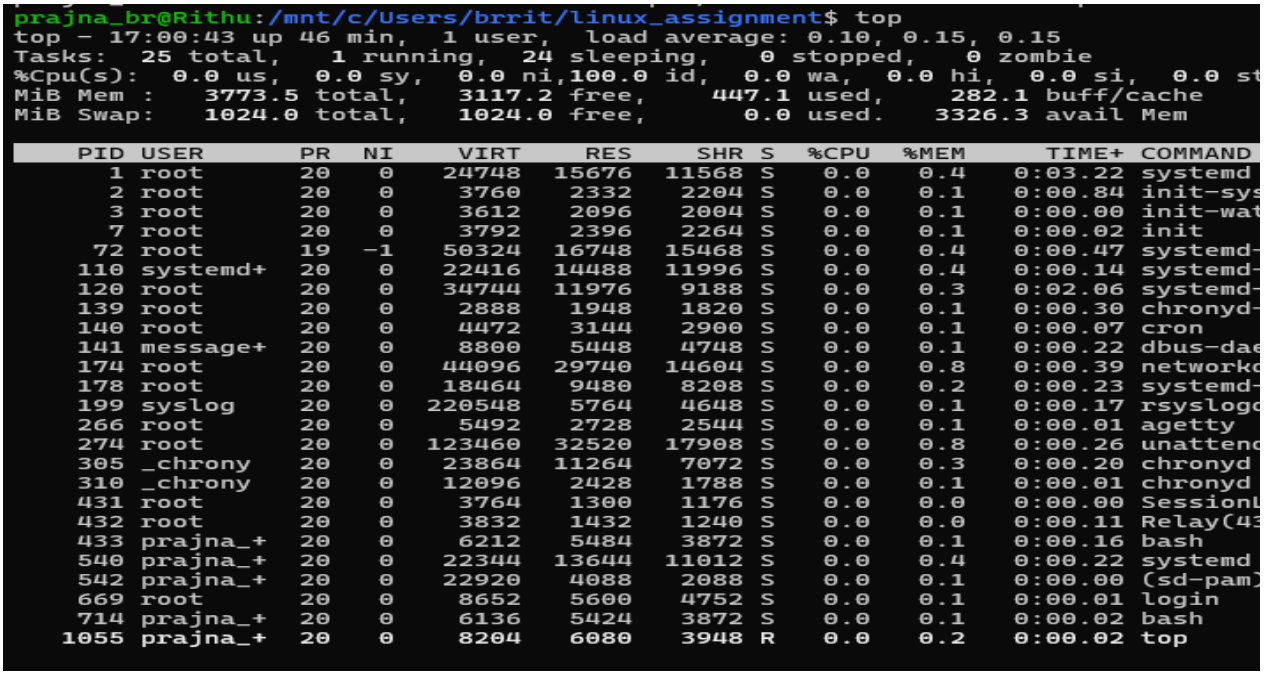
13. Check memory usage

```
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.7Gi       445Mi       3.0Gi       4.0Mi       281Mi       3.2Gi
Swap:          1.0Gi          0B       1.0Gi
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$
```

14.ps aux

```
thu:/mnt/c/Users/brrit/linux_assignment$ ps aux
 PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
   1  0.1  0.4  24748 15676 ?        Ss   16:14   0:03 /sbin/init
   2  0.0  0.0   3760  2332 hvc0     Sl+  16:14   0:00 /init
   3  0.0  0.0   3612  2096 hvc0     S+   16:14   0:00 init-watcher
   7  0.0  0.0   3792  2396 hvc0     Sl+  16:14   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
  72  0.0  0.4  50324 16748 ?        S<s  16:14   0:00 /usr/lib/systemd/systemd-journald
 110  0.0  0.3  22416 14488 ?        Ss   16:14   0:00 /usr/lib/systemd/systemd-resolved
 120  0.0  0.3  34744 11976 ?        Ss   16:14   0:02 /usr/lib/systemd/systemd-udevd
 139  0.0  0.0   2888  1948 ?        Ss   16:14   0:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
 140  0.0  0.0   4472  3144 ?        Ss   16:14   0:00 /usr/sbin/cron -f -P
 141  0.0  0.1   8800  5448 ?        Ss   16:14   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
 174  0.0  0.7  44096 29740 ?        Ss   16:14   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
 178  0.0  0.2  18464  9480 ?        Ss   16:14   0:00 /usr/lib/systemd/systemd-logind
 199  0.0  0.1 220548  5764 ?        Ssl  16:14   0:00 /usr/sbin/rsyslogd -n -iNONE
 266  0.0  0.0   5492  2728 tty1     Ss+  16:14   0:00 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/issue.d:/run/issue.d:/usr/lib/issue.d - linux
 274  0.0  0.8 123460 32520 ?        Ssl  16:14   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
 305  0.0  0.2  23864 11264 ?        S    16:14   0:00 /usr/sbin/chronyd -n -F 1 -x
 310  0.0  0.0  12096  2428 ?        S    16:14   0:00 /usr/sbin/chronyd -n -F 1 -x
 431  0.0  0.0   3764  1300 ?        Ss   16:14   0:00 /init
 432  0.0  0.0   3832  1432 ?        S    16:14   0:00 /init
 433  0.0  0.1   6212  5484 pts/0    Ss   16:14   0:00 /bin/bash
 540  0.0  0.3  22344 13644 ?        Ss   16:17   0:00 /usr/lib/systemd/systemd --user
 542  0.0  0.1  22920  4088 ?        S    16:17   0:00 (sd-pam)
 669  0.0  0.1   8652  5600 ?        Ss   16:17   0:00 login -- prajna_br
 714  0.0  0.1   6136  5424 pts/1    Ss+  16:17   0:00 -bash
1050 50.0  0.1   7160  4356 pts/0    R+   16:59   0:00 ps aux
thu:/mnt/c/Users/brrit/linux_assignment$
```

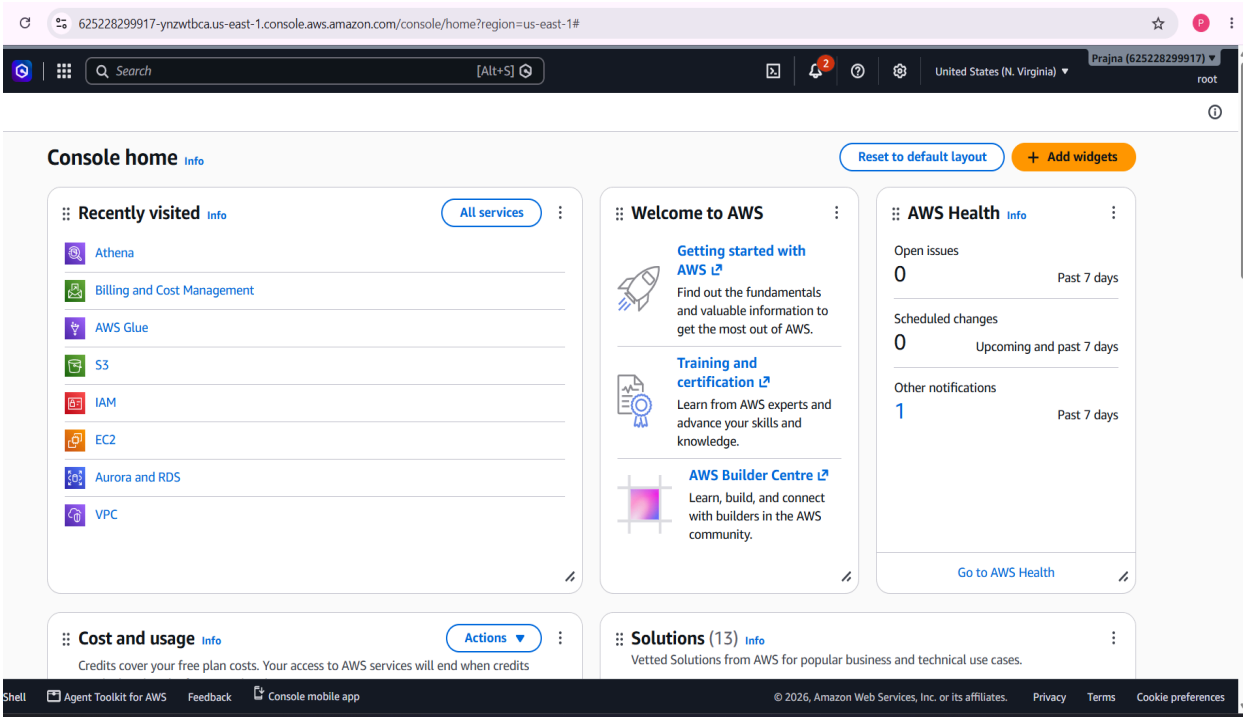
15.top



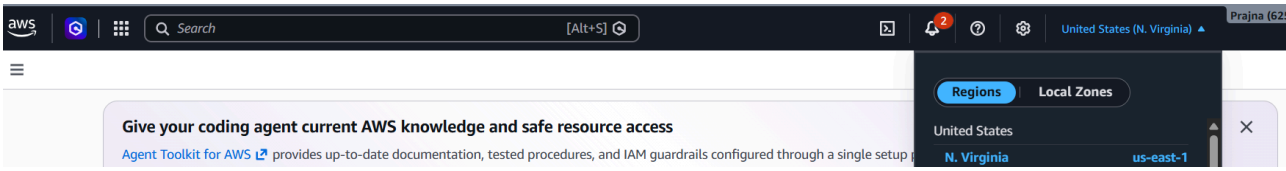
```
prajna_br@Rithu:/mnt/c/Users/brrit/linux_assignment$ top
top - 17:00:43 up 46 min,  1 user,  load average: 0.10, 0.15, 0.15
Tasks:  25 total,   1 running,  24 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3773.5 total,   3117.2 free,    447.1 used,    282.1 buff/cache
MiB Swap:   1024.0 total,   1024.0 free,      0.0 used.   3326.3 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   24748  15676  11568 S   0.0   0.4   0:03.22 systemd
      2 root      20   0    3760   2332   2204 S   0.0   0.1   0:00.84 init-sys
      3 root      20   0    3612   2096   2004 S   0.0   0.1   0:00.00 init-wat
      7 root      20   0    3792   2396   2264 S   0.0   0.1   0:00.02 init
     72 root      19  -1   50324  16748  15468 S   0.0   0.4   0:00.47 systemd-
    110 systemd+  20   0   22416  14488  11996 S   0.0   0.4   0:00.14 systemd-
    120 root      20   0   34744  11976   9188 S   0.0   0.3   0:02.06 systemd-
    139 root      20   0    2888   1948   1820 S   0.0   0.1   0:00.30 chronyd-
    140 root      20   0    4472   3144   2900 S   0.0   0.1   0:00.07 cron
    141 message+  20   0    8800   5448   4748 S   0.0   0.1   0:00.22 dbus-dae
    174 root      20   0   44096  29740  14604 S   0.0   0.8   0:00.39 networkd
    178 root      20   0   18464   9480   8208 S   0.0   0.2   0:00.23 systemd-
    199 syslog    20   0  220548   5764   4648 S   0.0   0.1   0:00.17 rsyslogd
    266 root      20   0    5492   2728   2544 S   0.0   0.1   0:00.01 agetty
    274 root      20   0  123460  32520  17908 S   0.0   0.8   0:00.26 unattend
    305 _chrony   20   0   23864  11264   7072 S   0.0   0.3   0:00.20 chronyd
    310 _chrony   20   0   12096   2428   1788 S   0.0   0.1   0:00.01 chronyd
    431 root      20   0    3764   1300   1176 S   0.0   0.0   0:00.00 SessionL
    432 root      20   0    3832   1432   1240 S   0.0   0.0   0:00.11 Relay(43
    433 prajna_+  20   0    6212   5484   3872 S   0.0   0.1   0:00.16 bash
    540 prajna_+  20   0   22344  13644  11012 S   0.0   0.4   0:00.22 systemd
    542 prajna_+  20   0   22920   4088   2088 S   0.0   0.1   0:00.00 (sd-pam)
    669 root      20   0    8652   5600   4752 S   0.0   0.1   0:00.01 login
    714 prajna_+  20   0    6136   5424   3872 S   0.0   0.1   0:00.02 bash
   1055 prajna_+  20   0    8204   6080   3948 R   0.0   0.2   0:00.02 top
```

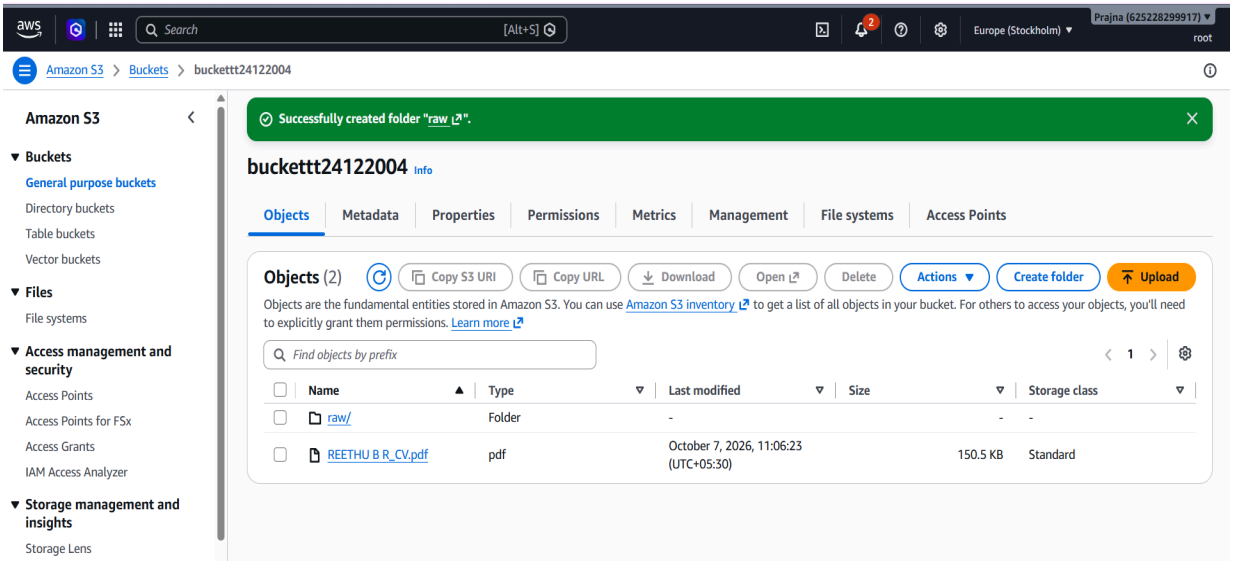
AWS Account / Region

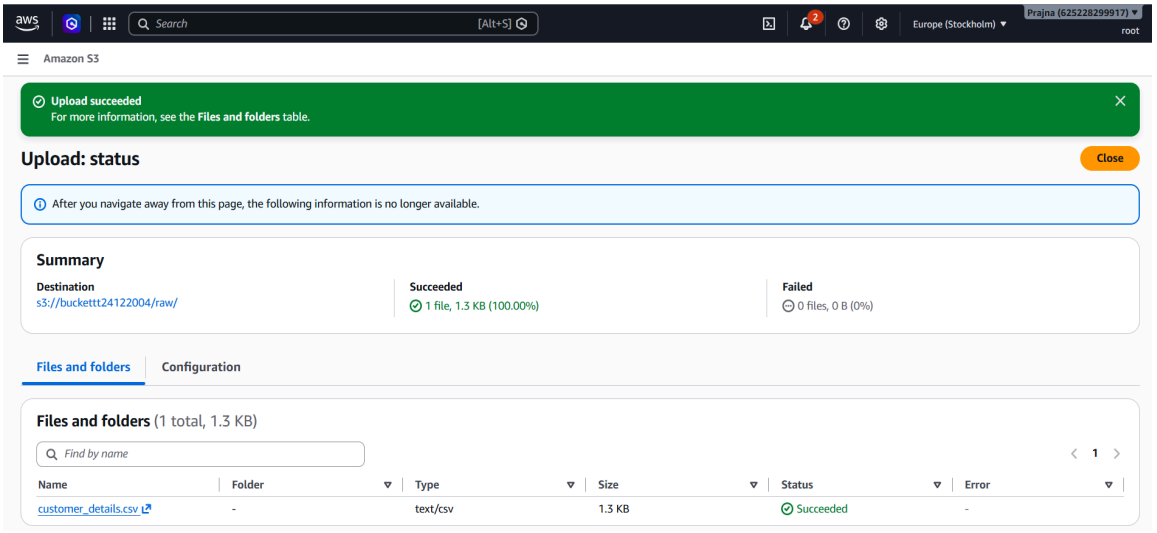


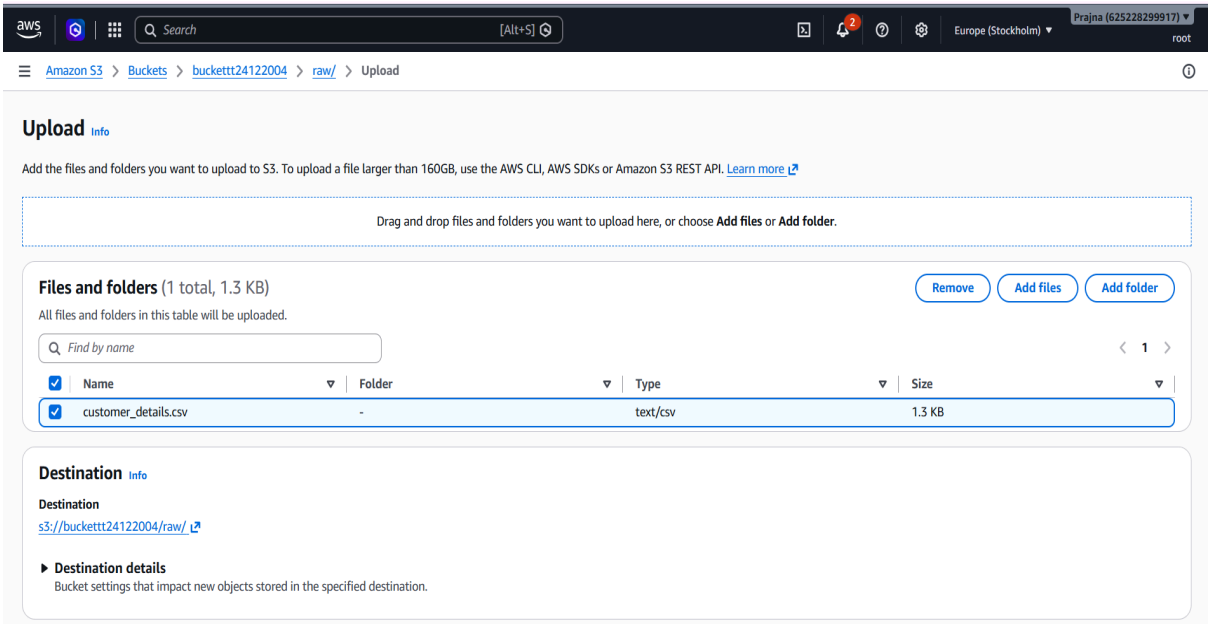
AWS Region : us-east-1



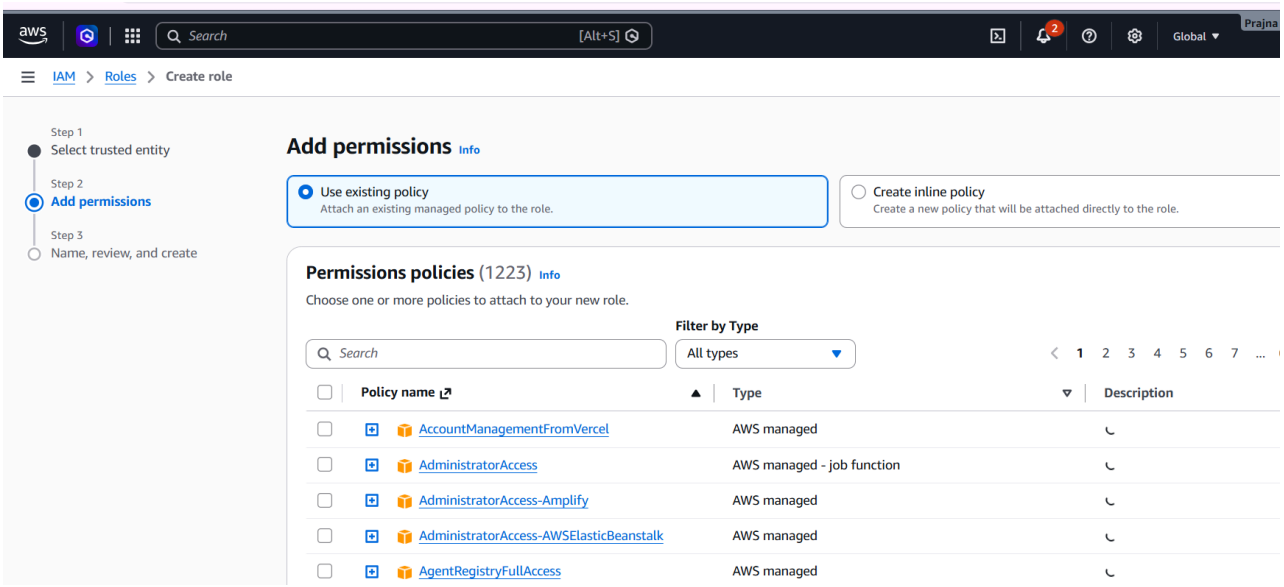
Amazon S3

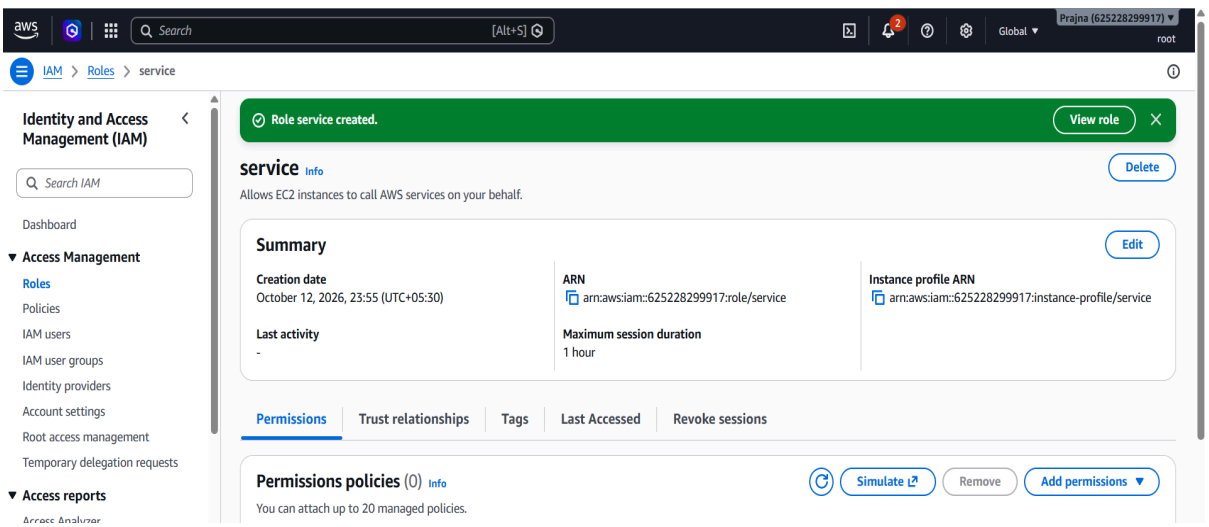

Search
[Alt+S]
Europe (Stockholm)
Prajna (625228299917)
root
Amazon S3 > Buckets > buckettt24122004
Amazon S3
Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets
Files
File systems
Access management and security
Access Points
Access Points for FSx
Access Grants
IAM Access Analyzer
Storage management and insights
Storage Lens
Successfully created folder "raw".
buckettt24122004 Info
Objects
Metadata
Properties
Permissions
Metrics
Management
File systems
Access Points
Objects (2)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
raw/
Folder
-
-
-
REETHU B R_CV.pdf
pdf
October 7, 2026, 11:06:23 (UTC+05:30)
150.5 KB
Standard


Search
[Alt+S]
Europe (Stockholm)
Prajna (625228299917)
root
Amazon S3
Upload succeeded
For more information, see the Files and folders table.
Upload: status
Close
After you navigate away from this page, the following information is no longer available.
Summary
Destination
s3://buckettt24122004/raw/
Succeeded
1 file, 1.3 KB (100.00%)
Failed
0 files, 0 B (0%)
Files and folders
Configuration
Files and folders (1 total, 1.3 KB)
Find by name
Name
Folder
Type
Size
Status
Error
customer_details.csv
-
text/csv
1.3 KB
Succeeded
-

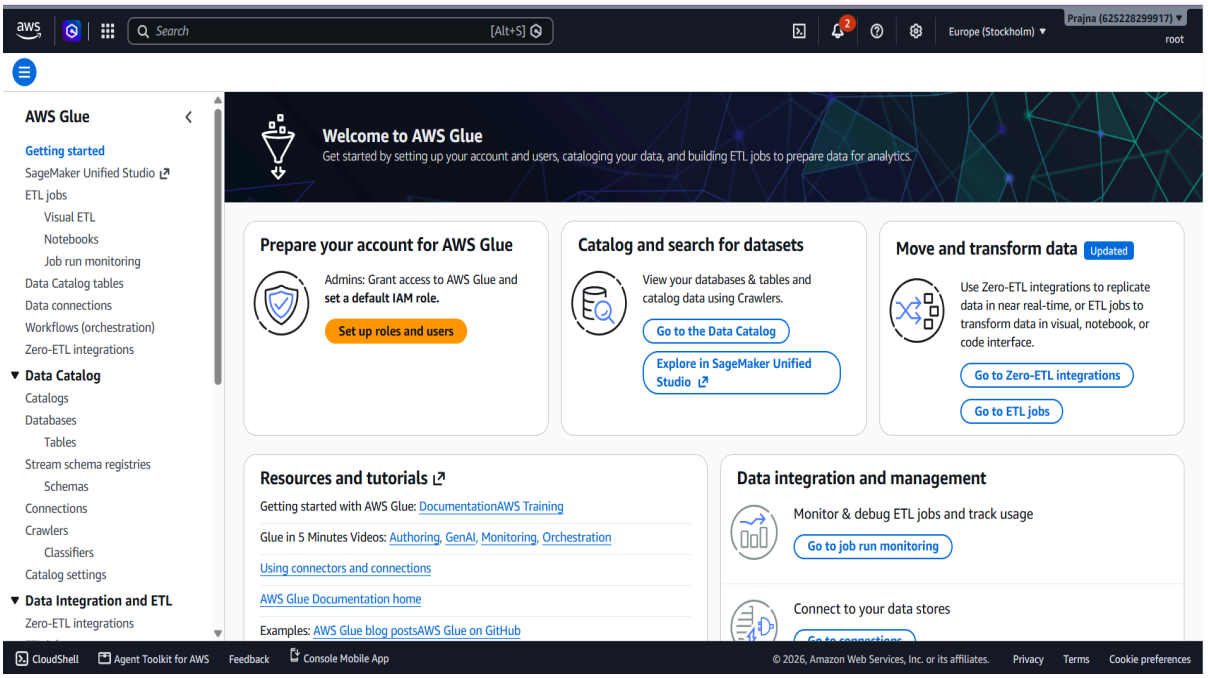


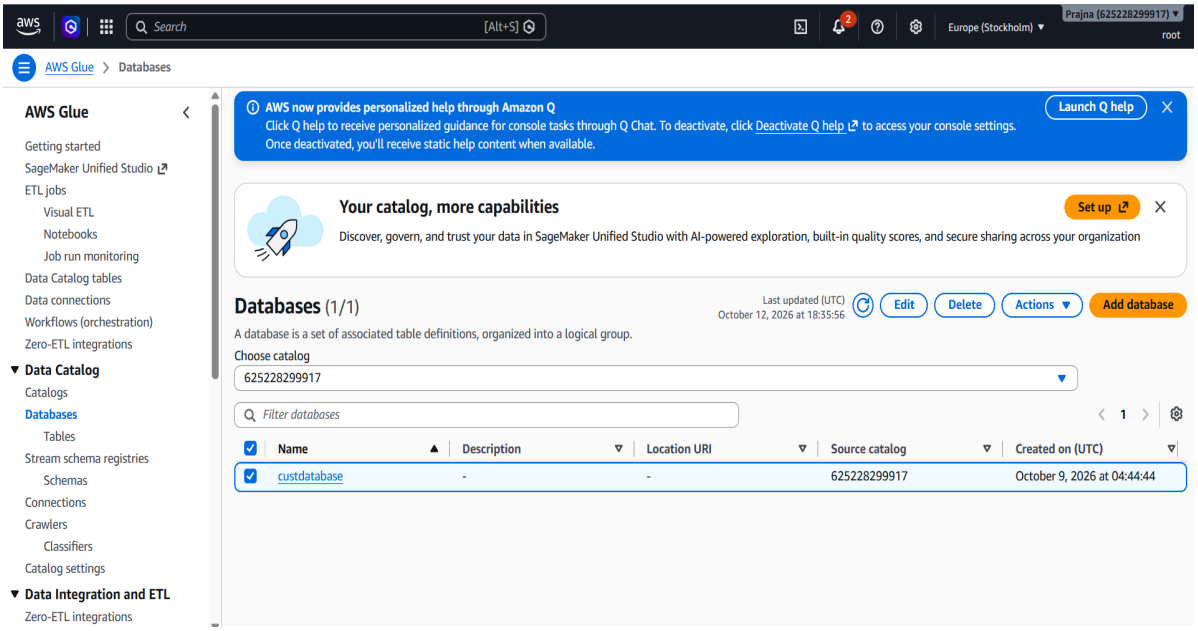
IAM (Identity and Access Management)

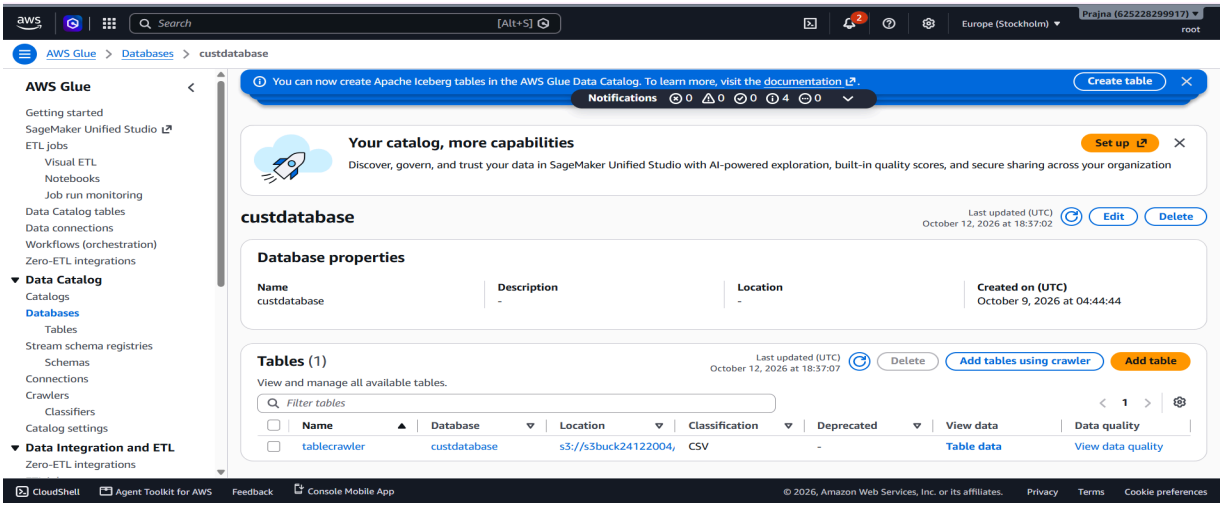




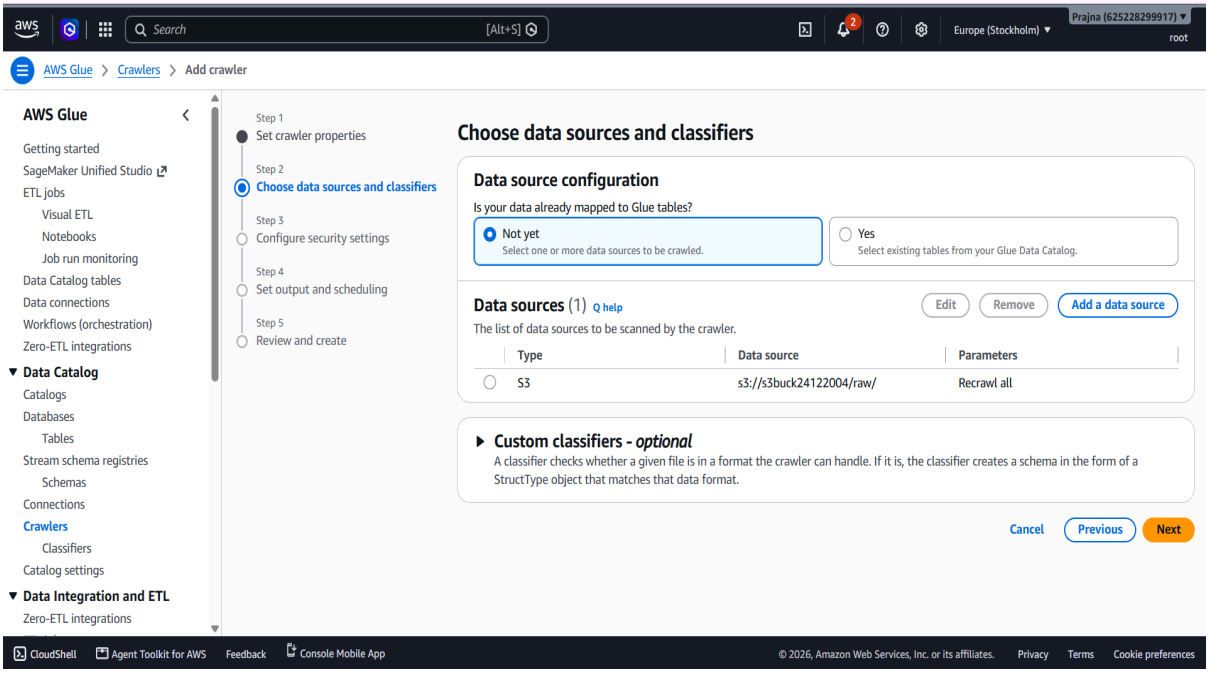
AWS Glue Database

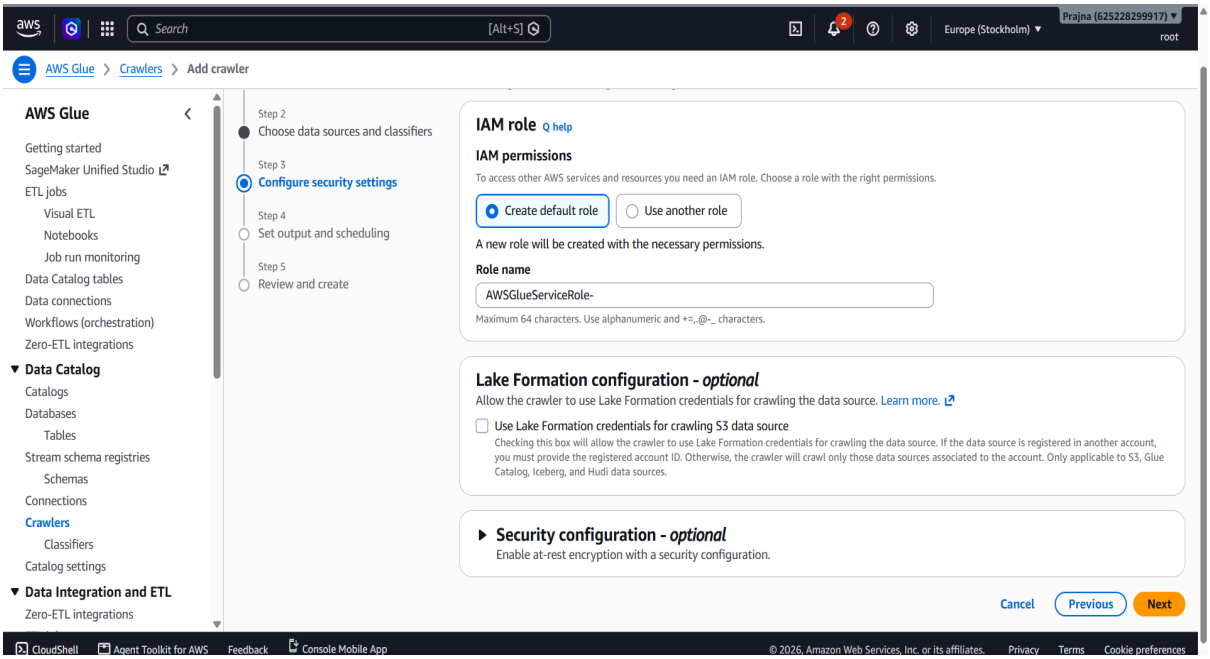


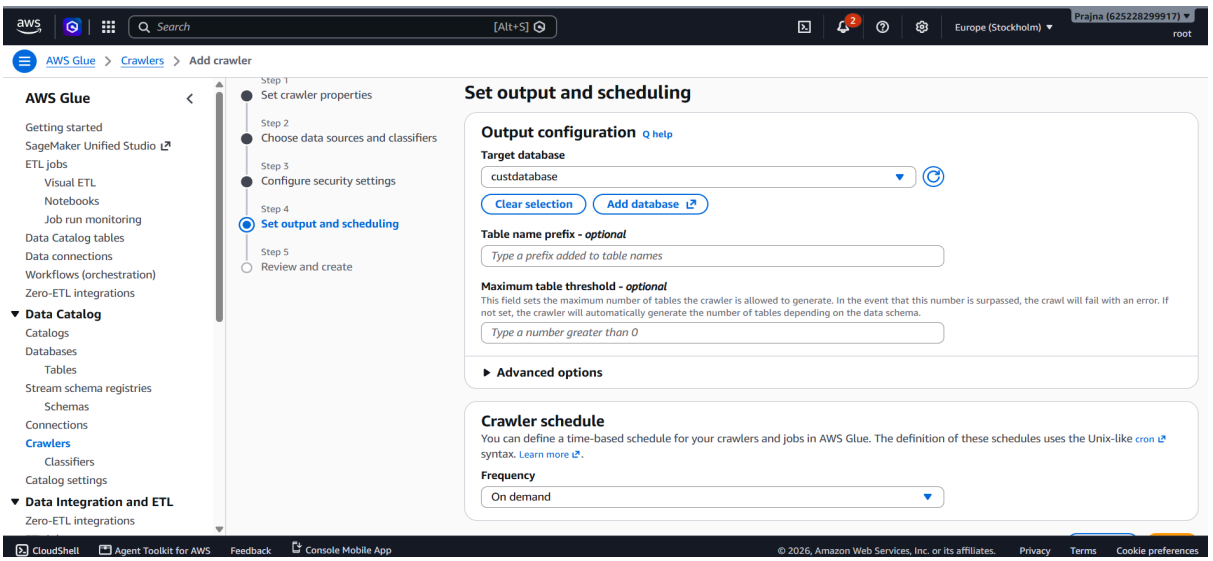


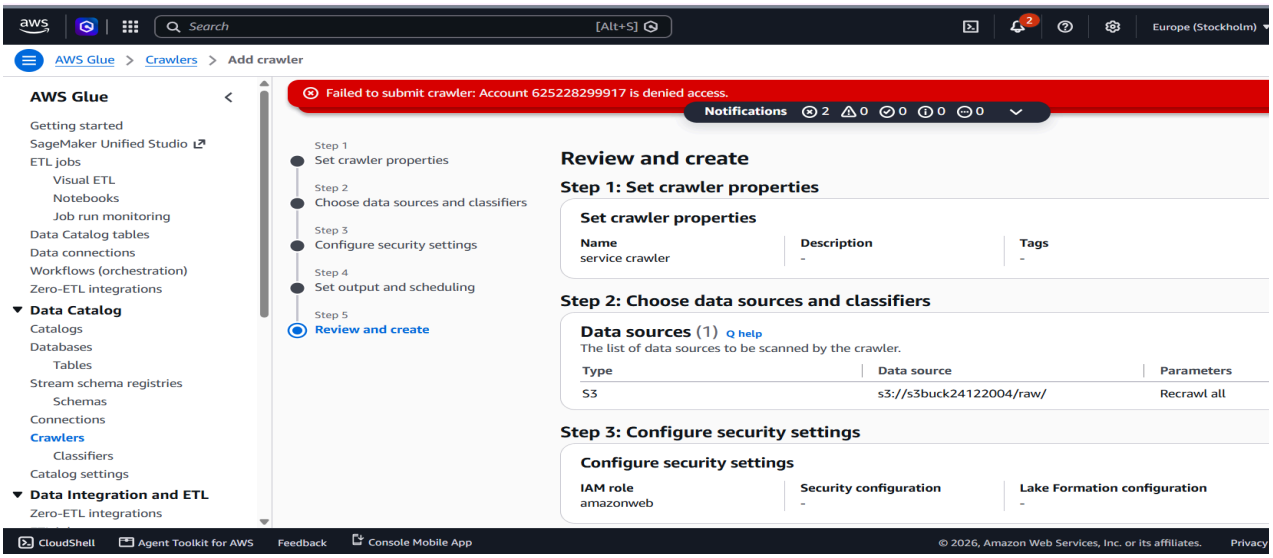


Glue Crawler

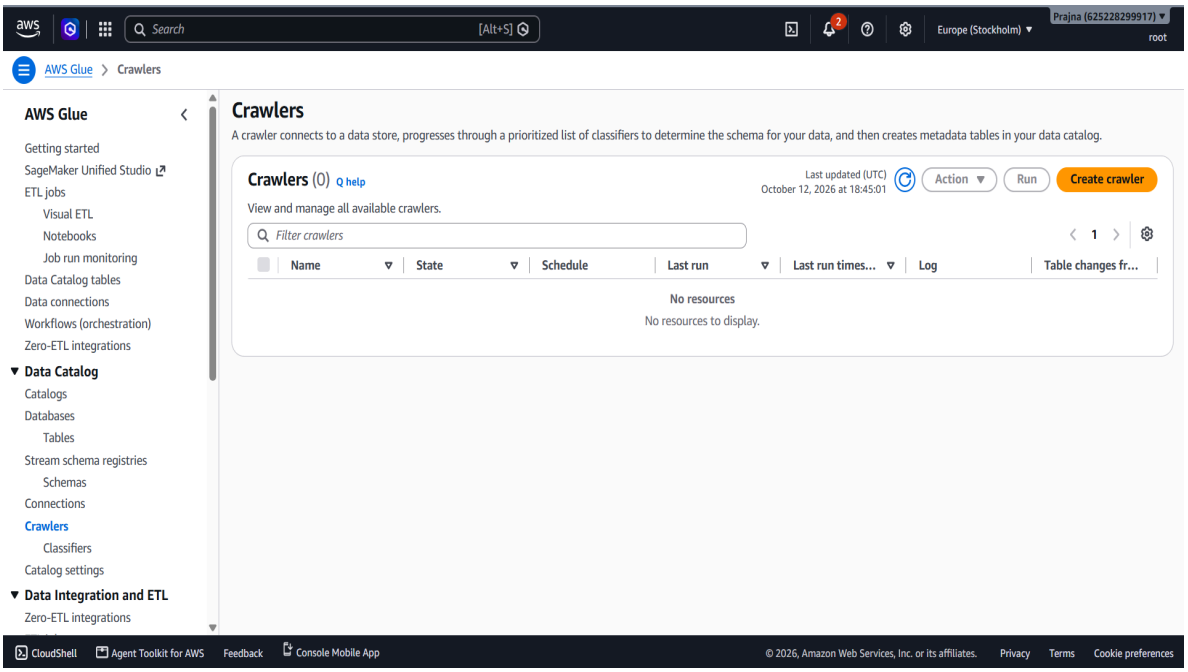




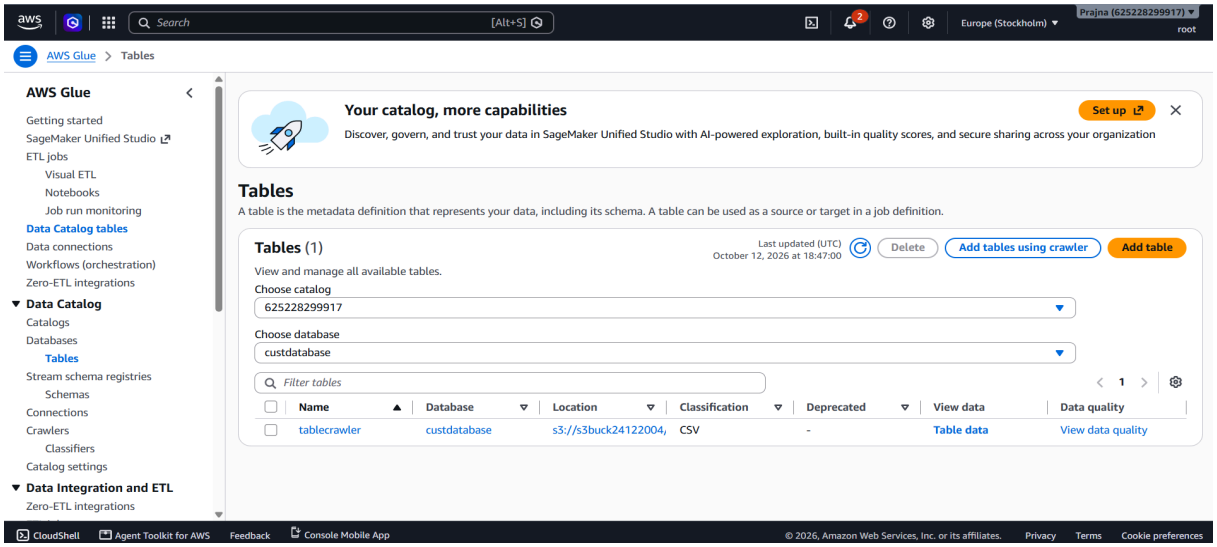


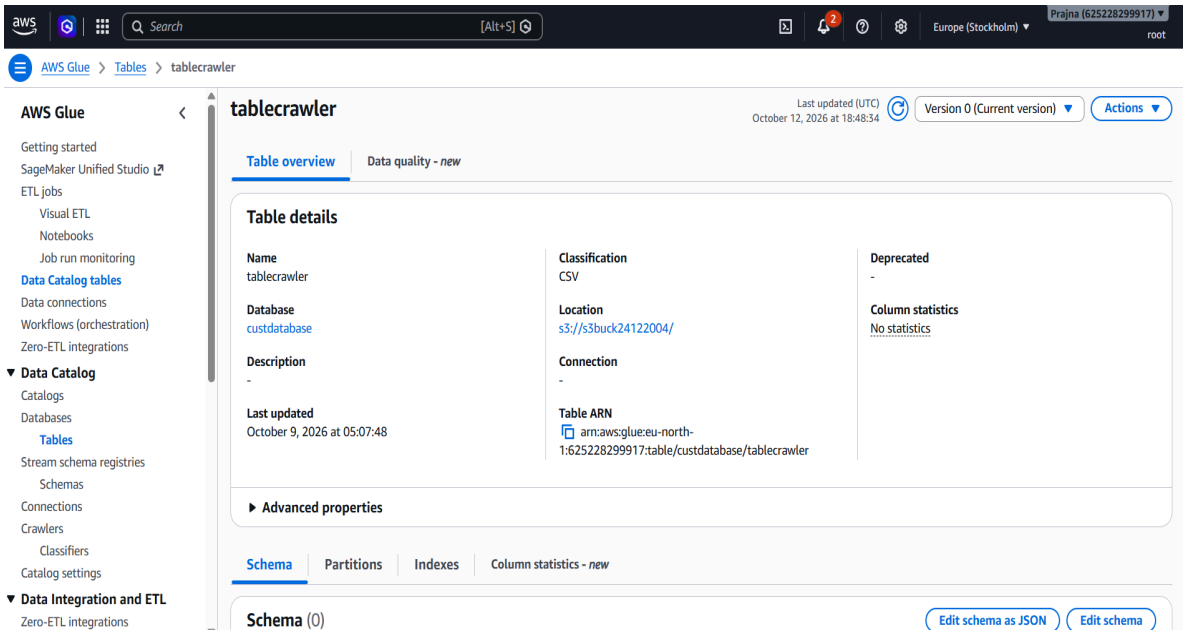


Crawler Execution

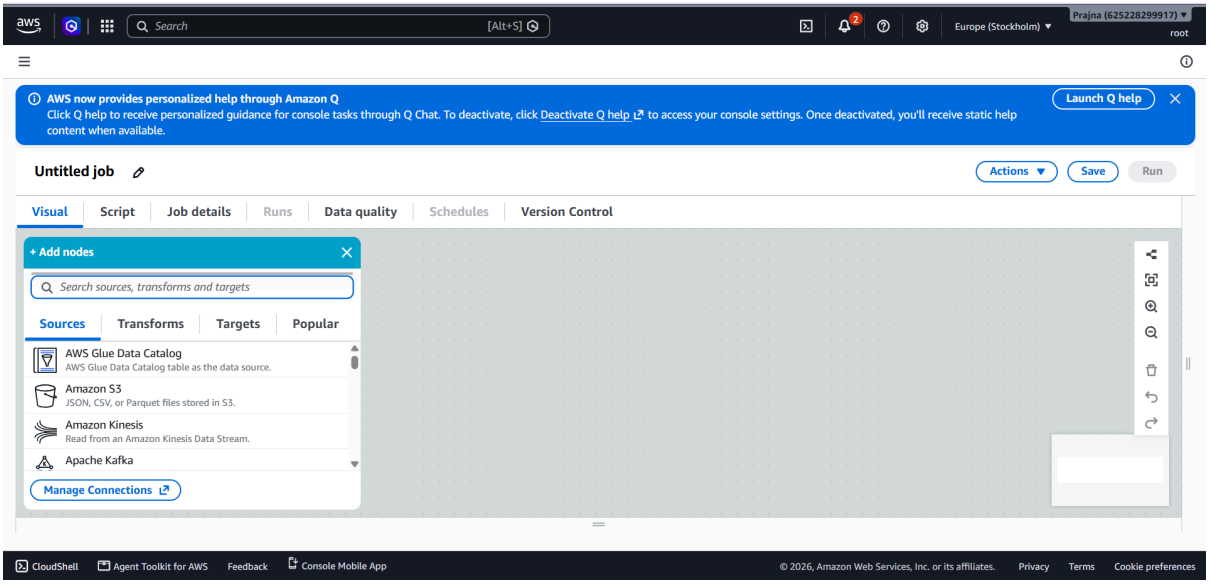


Glue Data Catalog

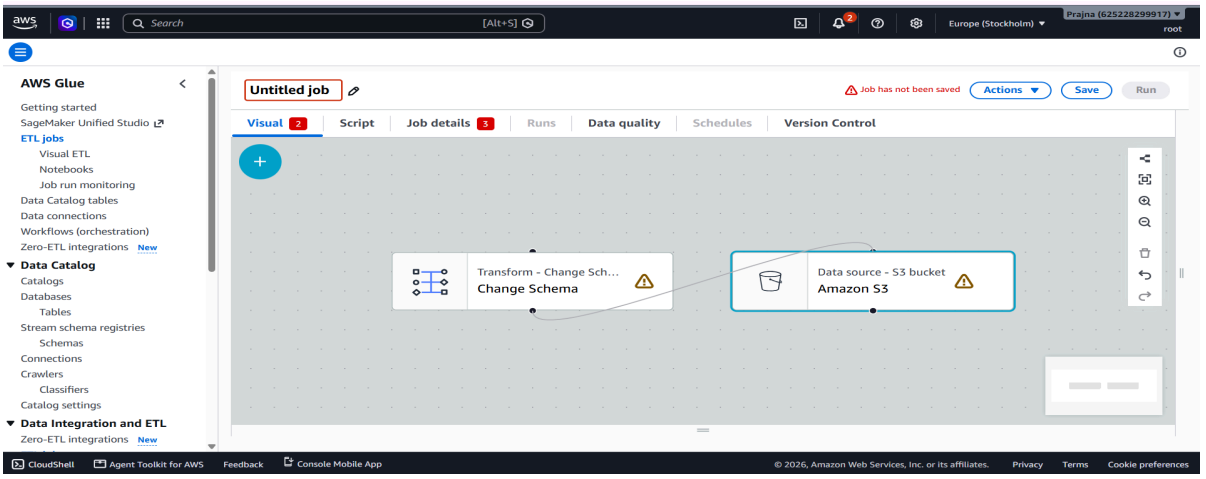




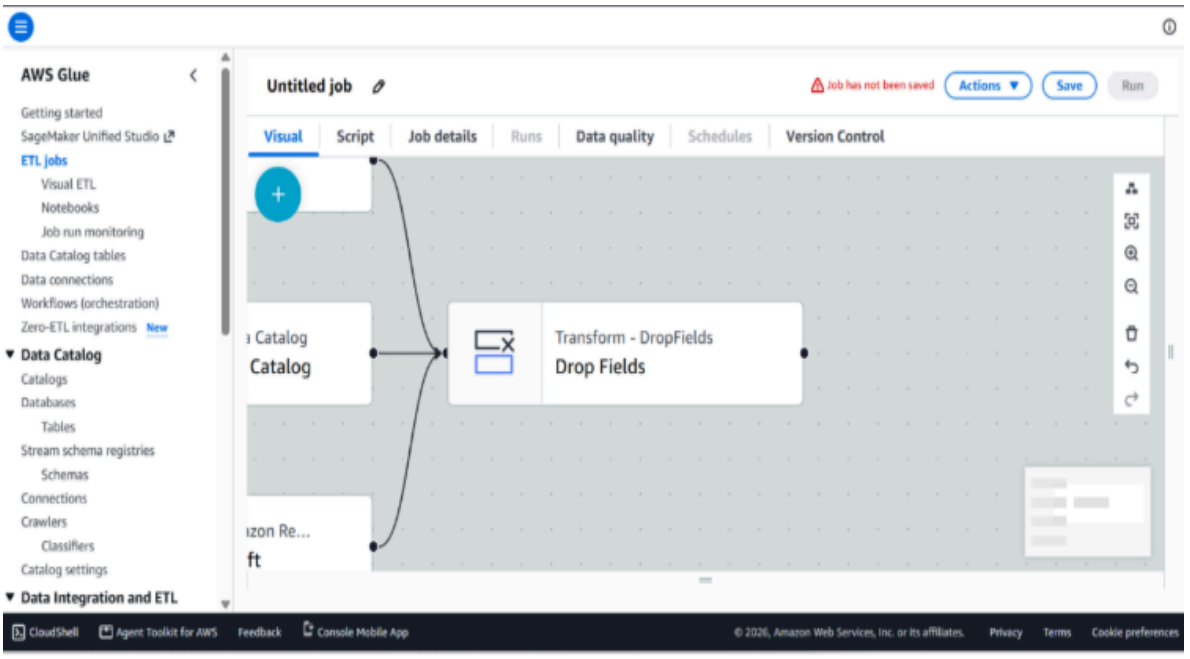
Glue Visual ETL



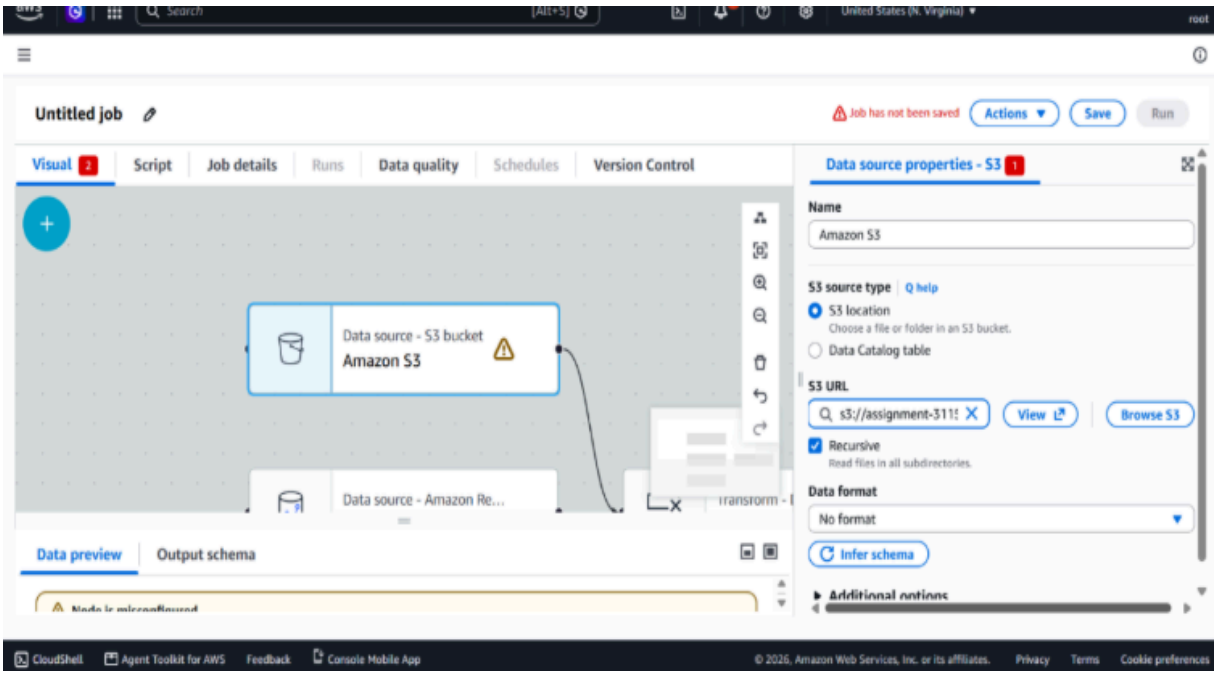
Glue Transformation

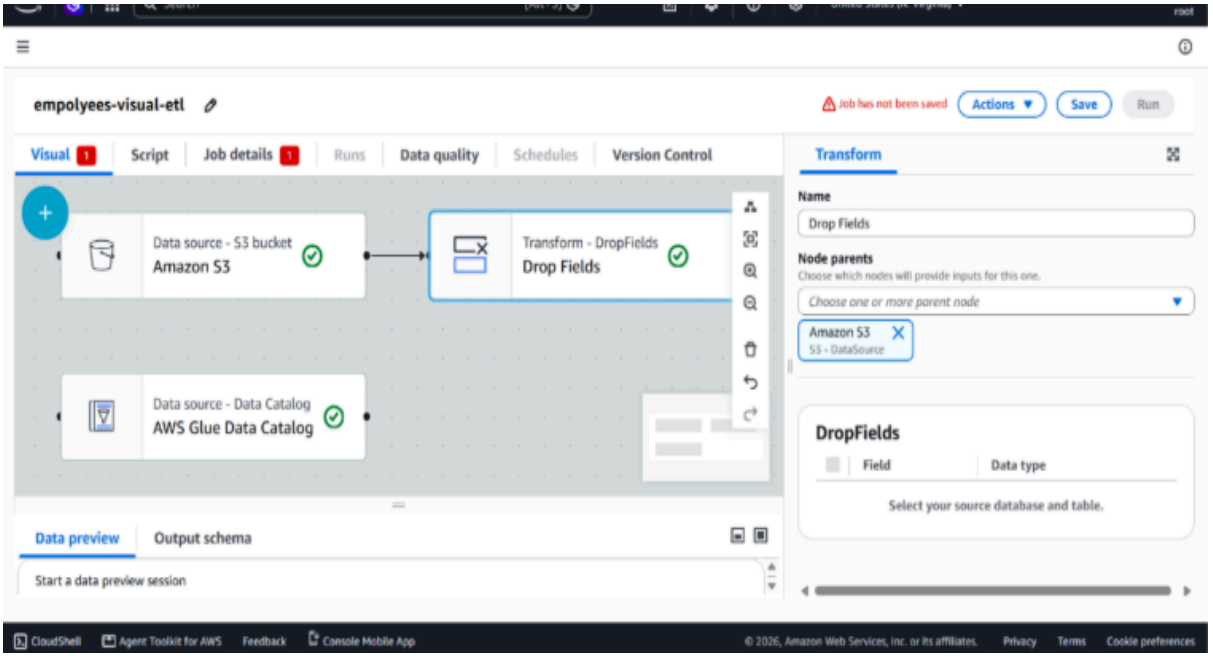


Drop fields

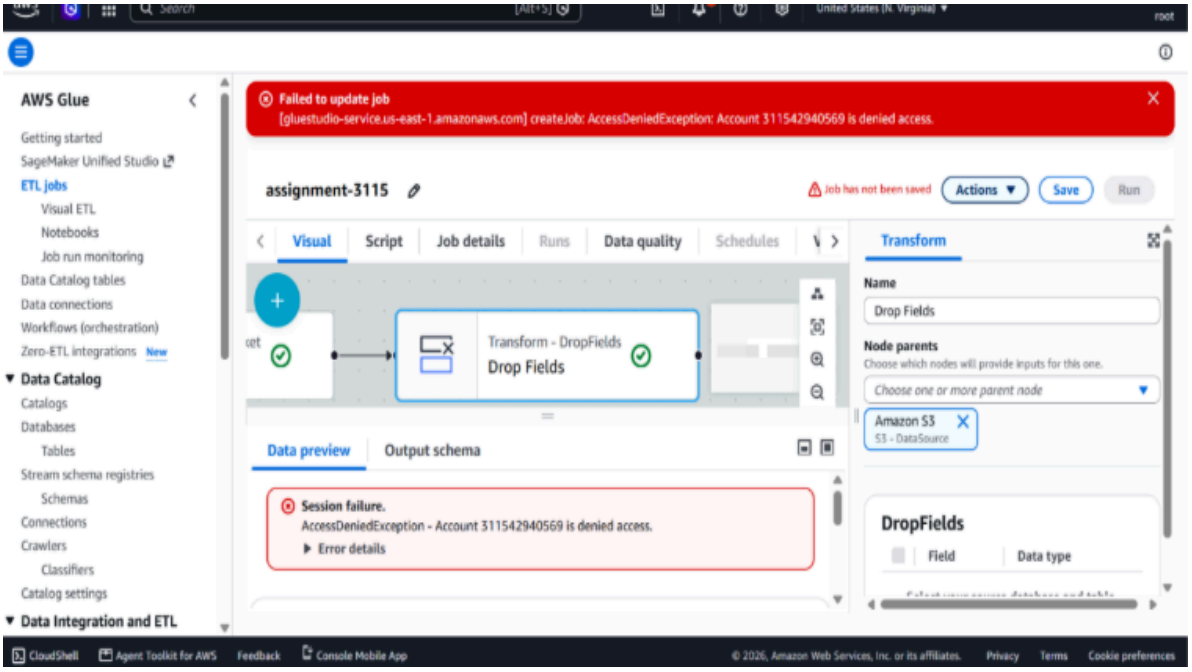


S3 Target Configuration





Job status = Succeeded/Denied



URBAN EDGE

Final Explanation.

1) Amazon S3:
Amazon S3 is used to store data files that are uploaded to an S3 bucket for further use.

2) AWS Glue Crawler:
The Glue Crawler scans data files & identifies schema, column names & data types.

3) Glue data catalog:
stores metadata & table information.

4) Glue Visual ETL:
Creates workflows to process source data.

5) Transformation:
Cleans data, removes duplicates & filters records.

6) Amazon S3 output:
stores the transformed data in separate folder.